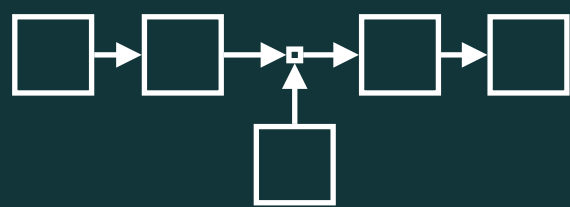




UNIVERSITY of
ROCHESTER
Department of Physics & Astronomy

$$\mathcal{H}[\sigma] = -\frac{1}{2} \sum_{ij} J_{ij} \sigma_i \sigma_j$$



$$\mathcal{T}_{Y \rightarrow X} = H[X_t | X \wedge Y_{t' < t}] \\ - H[X_t | X_{t' < t}]$$

$$\vec{v}_i(t + \Delta t) = \langle \vec{v}(t) \rangle_R$$

$$\vec{x}_i(t + \Delta t) = \vec{x}_i(t) + \vec{v}_i(t) \Delta t$$

$$\Delta S \geq 0$$



INFORMATION- DRIVEN STATES OF MATTER

SUMMER WORKSHOP

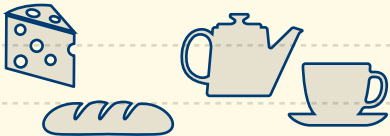
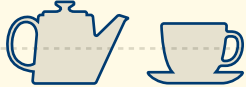
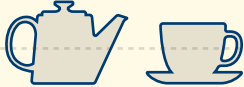
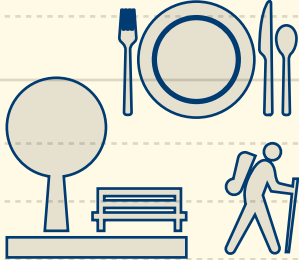
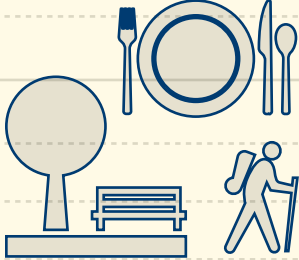
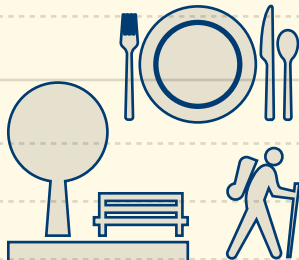
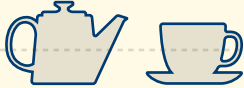
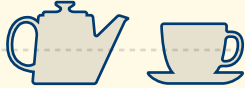
JULY 10-12, 2024



JOHN
TEMPLETON
FOUNDATION

Inspiring Awe & Wonder

SCHEDULE

	WEDNESDAY, JULY 10	THURSDAY, JULY 11	FRIDAY, JULY 12
8:00 AM			
9:00 AM	OPENING REMARKS		
	THE MEANING OF LIFE: TOWARDS AN EFFECTIVE THEORY OF SEMANTIC INFORMATION DAMIAN SOWINSKI	THE SEARCH FOR INFORMATIONAL SYSTEMS IN THE UNIVERSE CALEB SCHARF	MOLECULAR COMPUTING WITH CHEMISTRY JUAN PEREZ MERCADER
10:00 AM	AVOIDING THE AD HOC ASCRPTION OF DEATH MATTHEW EGBERT	AN INFORMATIONAL PERSPECTIVE OF LIFE MICHAEL WONG	INFORMATIONAL STEADY- STATES AND JUMP PATTERNS IN CONTINUOUSLY MONITORED QUANTUM SYSTEMS GABRIEL LANDI
11:00 AM			
NOON	WHAT DOES IT ALL MEAN? WHEN IS INFORMATION HUNGRY FOR CAUSAL INFLUENCE? STUART BARTLETT	INFORMATION SENSING AND TRANSMISSION IN ASTROBIOLOGICAL ENVIRONMENTS: CONSTRAINTS ON LIVING SYSTEMS AND BIOSIGNATURES MANASVI LINGAM	NON-EQUILIBRIUM STATISTICAL PHYSICS OF COMPUTATION AND COMMUNICATION DAVID WOLPERT
1:00 PM			
2:00 PM	ACTIVE OSCILLATING MATTER: SWARMS OF SWIMMING NEMATODES AND INTERACTING OSCILLATOR MODELS FOR THEIR COLLECTIVE WAVES ALICE QUILLEN	FUNDAMENTAL COSTS OF PREDICTION JENNY POULTON	OPTIMAL SINGLE-LETTER CODES, KELLY BETS, AND BIOLOGICAL INFORMATION PROCESSING ANDREW ECKFORD
3:00 PM			
4:00 PM	INFORMATION PROCESSING IN BACTERIAL CHEMOTAXIS: DISSIPATION-ENHANCED COOPERATIVE SENSING AND ADAPTATION DAVID HATHCOCK	HOW WELL DO NEURONS, HUMANS, AND ARTIFICIAL NEURAL NETWORKS PREDICT? SARAH MARZEN	INFORMATION-THEORETIC LIMITS ON GROWTH IN MINIMAL CHEMICAL SYSTEMS JORDI PIÑERO
5:00 PM	PREDICTING THE FUNCTION OF BIOMOLECULES WITH INFORMATION THEORY: NO TRAINING OR MODELING REQUIRED CHRIS ADAMI		CLOSING REMARKS

*Breakfast, Coffee, and Lunch provided by the
University of Rochester*

Dear Participant,

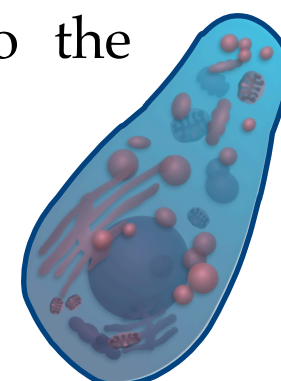
One of the most distinguishing characteristics of life is its ability to harness flows of information for the purposes of self-creation and self-maintenance. What makes life different? Can this difference ever be replicated in information-powered machines, which are starting to be studied both theoretically and experimentally? And what are the new physics, methods, and concepts required to understand it? These are the kinds of questions we hope to discuss at this workshop entitled "Information-Driven States of Matter".

In organizing the workshop, we wanted to recognize the central role of information in distinguishing the usual domains of "dead" physics and the self-organized, far-from-equilibrium dynamics embodied by living systems, their near antecedents and their possible machine descendants.

Our meeting will engage with many distinct but interrelated issues. These include astrobiological questions regarding information-processing in biospheres and the necessity of identifying life's "computational habitable zones". Other discussions will touch upon fundamental issues in the physics of information, ranging from the statistical mechanics of computation to the role of information in quantum systems. The nature of information processing in living and artificial systems will also be explored, ranging from ultimate bounds on prediction to the function of information in chemotactic organisms and other kinds of active matter. The operational meaning of information will be explored via discussions of Kelly betting, semantic information, as well as their links to basic questions of a systems' viability.

Because this is a workshop, we are hopeful that our meeting will stimulate vibrant and useful discussions. For that reason we have set the time for talks to 40 minutes, leaving 20 minutes after each talk for questions, comments and expansions. We are hopeful these periods will be enough to generate themes whose exploration can be continued during meals and over libations.

We thank you so very much for attending and we look forward to the workshop.



ORGANIZERS

This workshop is funded by a grant from the John Templeton Foundation, titled "Information Architectures That Enable Life" (#62417).



Adam Frank is an astrophysicist whose work has included the magnetohydrodynamics of star formation and stellar death as well as the evolution of exoplanet atmospheres. His recent work has focused on astrobiology with an emphasis on the long-term thermodynamic coupling of coupled planetary systems with both bio and technospheres. It is the study of life in its planetary context that brought Frank to questions of information-based narratives for agency and autonomy. This has led to work on the development of operational theories of semantic information.



Gourab Ghoshal is a theoretical physicist who works in the field of Complex Systems writ large. His research interests are in the theory and applications of Complex Networks as well as Non-equilibrium Statistical Physics, Game theory, Econophysics, Dynamical Systems and the Origins of Life. His recent work has focused on information architectures and their role in complex adaptive systems.



Artemy Kolchinsky works at the intersection of complex systems, information theory, and nonequilibrium thermodynamics. He is particularly interested in nonequilibrium systems that are able to acquire information about their environments and use it to maintain their own existence. Are there general laws that govern such systems? Can these systems help us understand the origin of life and mind?

*Predicting the function of biomolecules with information theory:
no training or modeling required*

CHRIS ADAMI



Understanding and predicting the function of biomolecules occupies a central role in all of biology because these are the building blocks that all of life is made out of. The function of biologically active molecules such as proteins and RNAs is determined by the information stored in the molecule's sequence, but as yet we have not learned how to efficiently access this information. I describe a new approach to function prediction that is rooted entirely in the theory of information, by taking advantage of Fano's information decomposition theorem of 1961. I show that this method makes it possible to quantify the information in a single sequence that is predictive of the function, without any training or modeling, using only multiple-sequence alignments (MSAs) of example sequences. In theory, this method should beat the accuracy of any and all competing methods (including Deep Learning), because it extracts only information, leaving behind the noise. I show examples where this is true in practice.

What does it all Mean? When is Information Hungry for Causal Influence?

I will discuss what I view as promising avenues for observing 'information-driven' transitions in non-living systems. I will first present work on endowing simple reaction-diffusion structures with rudimentary learning abilities, and discuss the implications of these properties for the emergence of semantic information. I will then suggest a potential avenue for an informational quantity closely related to the semantic information of Kolchinsky & Wolpert (2018). To close, I will discuss the propositions of: 1) whether the Free Energy Principle due to Friston et al. has the power to formalize information-driven transitions, and 2) whether the origins of life should be considered an information-drive transition.

**STUART
BARTLETT**



Optimal single-letter codes, Kelly bets, and biological information processing

**ANDREW
ECKFORD**



We show that an optimal single-letter source-channel code is also a Kelly bet, i.e., there exists an investment game with noisy information for which it gives the optimal expected log growth rate. For the converse problem, for any investment game in which the log growth rate of fitness is the figure of merit, we show that the rate-distortion tradeoff has a simple linear bound, and that the bound is achievable at the point where the corresponding single-letter code is optimal. This interpretation has useful analytic consequences: it is possible to find investment games in which optimal single-letter codes exist at zero, one, and a continuum of points on the rate-distortion curve. Using this approach, the advantage of adding actions and strategies can be explicitly bounded. Finally, we discuss the implications of this result for situations where log growth rate is used, especially in biological decision problems.

Avoiding the Ad Hoc Ascription of Death

The idea of homeostasis—seeking or maintaining some steady state or equilibrium—has been invoked to explain key aspects of cognition, such as agency and adaptation. In models used to explore these ideas, a common strategy is to designate a 'death' attractor that the system is to avoid if it is to 'survive'. When this attractor is avoided, the system is seen as having demonstrated adaptation or agency or some other aspect of cognition. But the choice of which attractor is to be avoided is often only an arbitrary ascription. The death attractor is bad only because the researcher exploring the model says so. In these models there is often no actual agent in the sense that there is no entity that can die (i.e., break irrevocably or cease to exist). Such models can be useful in understanding how a system can be smart and solve problems. But if we want to understand how an agent might have problems of its own and act so as to solve those problems, we need to use models that don't abstract away the details of being an agent. We need models in which an agent can be distinguished from its environment not by ascription but by principle and where dynamics are good or bad for that agent because of the way that they affect that agent's emergent (not prescribed) viability. I will review a few computational models that work in this way. In these models, the agents have a common underlying organisation. Each consists of a collection of interdependent components, where each component depends upon and enables other components in the network. As I'll describe, this organisation implies a set of needs that are intrinsic to the agent (i.e., not arbitrarily ascribed by a 3rd party observer). I'll use these models to explore how a system can respond to its own dynamic and context-dependent needs—an essential aspect of agency.

**MATTHEW
EGBERT**



Information processing in bacterial chemotaxis: dissipation-enhanced cooperative sensing and adaptation

Chemotaxis signaling pathways in *Escherichia coli* are driven out of equilibrium by ATP hydrolysis, which enables the phosphorylation of messenger proteins that carry signals from the chemoreceptors on the cell surface to the motors that control the bacteria's swimming motion. I will discuss two recent measurements of chemosensory response and dynamics that reveal subtle, but clear signatures of this underlying dissipation. First, adaptation by methylation of chemoreceptors shifts the downstream intracellular response by two orders of magnitude in ligand concentration, which is disproportionately larger than the adaptive shifts in ligand binding response. Second, even in the absence of external stimuli, cells spontaneously switch between active and inactive kinase states, with dynamics that break time-reversal symmetry. To understand these behaviors, we introduce nonequilibrium allosteric and lattice models of the chemoreceptor array that resolve these experimental observations and explain they how arise from dissipative collective dynamics. Our models also provide new insights into the microscopic mechanisms underlying of the chemosensory array and the role of energy dissipation in enhancing cooperative sensing and adaptation.

**DAVID
HATHCOCK**



*Information Sensing and Transmission in Astrobiological Environments:
Constraints on Living Systems and Biosignatures*

**MANASVI
LINGAM**



Envisioning and searching for diverse forms of life warrants an agnostic approach grounded in the principles of (bio)physics and (bio)chemistry. The capacity to sense and transmit information constitutes a ubiquitous feature of life on Earth, and is plausibly a universal attribute of living systems. In this work, the constraints arising from the ambient environment vis-à-vis information sensing and transmission are delineated, and the ramifications of these constraints for (extra)terrestrial life are analyzed. Achieving effective information sensing by detecting gradients translates to a lower bound on the cell size that is only weakly dependent on the physicochemical properties of the habitat (such as temperature and viscosity), thereby suggesting a certain degree of generality. This size limit might therefore serve as a means of falsifying or corroborating potential microfossils recovered from a given setting, complementing existing frameworks. In contrast, information transfer via molecular communication is sensitive to a variety of environmental characteristics, indicating that certain environments (e.g., hydrocarbon lakes on Titan, Saturn's moon) are more conducive than others. Hence, such settings might have facilitated the early evolution of (proto)life, because information sharing can confer many evolutionary benefits on microbes.

*Informational steady-states and jump patterns in
continuously monitored quantum systems*

A central feature of quantum mechanical systems is the information-back action trade-off: measurements necessarily have a disturbing effect and the more information one inquires, the stronger is the disturbance. In this talk I will discuss our recent results characterizing these effects in steady-states of continuously monitored quantum systems. First I will discuss experimental results that allow us to assess the entropy production (a measure of irreversibility) in said systems, and the role that information acquisition has in modifying the steady-state. Afterwards I will move on to discuss how continuous measurement data can be used to try to infer the stochastic dynamics of a quantum state in Hilbert space, with a particular emphasis on the fundamental differences with hidden Markov models. Finally, I will discuss the relevance of these findings in understanding the thermodynamics of time-keeping.

**GABRIEL
LANDI**



How well do neurons, humans, and artificial neural networks predict?

Sensory prediction is thought to be vital to organisms, but few studies have tested how well organisms and parts of organisms efficiently predict their sensory input in an information-theoretic sense. In this talk, we report results on how well cultured neurons ("brain in a dish") and humans efficiently predict artificial stimuli. We find that both are efficient predictors of their artificial input. That leads to the question of why, and to answer this, we study artificial neural networks, finding that LSTMs show similarly efficient prediction but do not model how humans learn well. Instead, it appears that an existing model of cultured neurons and a model of humans as order-R Markov modelers explain their performance on these prediction tasks.

**SARAH
MARZEN**



Molecular Computing with Chemistry

Living systems compute using biochemistry. Chemical reactions can be thought as the ultimate materially implemented information recognition machines. Thinking of chemical reactions as molecular automata, the full range of computing automata in the Chomsky inclusive hierarchy can be realized in one-pot, table-top chemical reactors without using biochemistry. From the simplest, Finite automata using a precipitation reaction, to the most complex, the Turing machine, which requires non-linear chemistry. Chemical automata involve an Avogadro's number of processors and their efficiency can be optimized to a very high degree. They have both digital and analog features and their architecture is not von Neumann. Language acceptance/rejection criteria by these automata consist of specific molecular profiles, (including radicals) that can be directly used in-situ or be chemically processed downstream at the molecular and/or supramolecular levels. Acceptance implies the minimization of paths for an action-like quantity. We will illustrate the above by the construction and operation of a Turing machine in the linear bounded automaton class (LBA) using the Belousov-Zhabotinsky (BZ) chemical reaction.

**JUAN PEREZ
MERCADER**



Information-theoretic limits on growth in minimal chemical systems

We establish an information-theoretic limit to growth rates in simple chemical reaction systems, focusing on autocatalytic networks within periodically fluctuating environments. We introduce an empirically accessible measure of information that: (1) is realistically achievable in a finite reactor; (2) its principal control parameters are experimentally accessible; and (3) operates independently of any predefined sensory mechanisms, thus representing intrinsic information processing at a fundamental form. Our analysis examines how average chemical yields adjust to environmental fluctuations that favor different autocatalytic species in the network. We propose the concept “information gap”, a new metric that complies with (1-3). This operational measure allows us to study different memory regimes of the network, argue about the system’s semantic information interpretation and consider its relevance in the context of prebiotic life.

**JORDI
PIÑERO**



Fundamental costs of prediction

The ability to predict the future is beneficial to many biological systems. For example, some organisms can react to an environment without glucose by making a costly adaptation to allow processing different types of sugars such as xylose. However, this is only worth doing if the cell can predict that glucose is unlikely to return. Alternatively, in bacterial chemotaxis, the cell must detect the gradient of resources it has experienced to predict whether an area of high resource density lies ahead. To make good predictions, a system must collect information about the current and past state of the environment and this information gathering also costs resources. In this talk we calculate the fundamental limits on the cost of extracting various useful pieces of information about a given environment and consider simple systems able to reach these limits. We discuss how a system can exploit correlations within the environment to reduce the costs of information gathering, and the consequences of the system having made an incorrect guess about the statistics of the environment

**JENNY
POULTON**



Active oscillating matter: Swarms of swimming nematodes and interacting oscillator models for their collective waves

Vinegar eels, a type of swimming nematode, collectively organize to form a traveling wave that drives fluid circulation. Vinegar eels are a biological system in the class of active matter called *swarmalators* --- which are interacting agents that self-propel and can collectively synchronize. Models of an interacting chain of oscillators can exhibit wave-like phenomena, similar to that seen in concentrations of vinegar eels. For locomotion and transport of fluids, biological systems can rely on a collective traveling wave. However, many models of interacting oscillators do not robustly or uniquely converge onto wave-like behavior. This leads us to ask: what are the properties of biological systems that make them robust to wave formation? Biological systems may leverage stochastic phenomena, lopsided interactions or variations in oscillator properties.

**ALICE
QUILLEN**



The Search for Informational Systems in the Universe

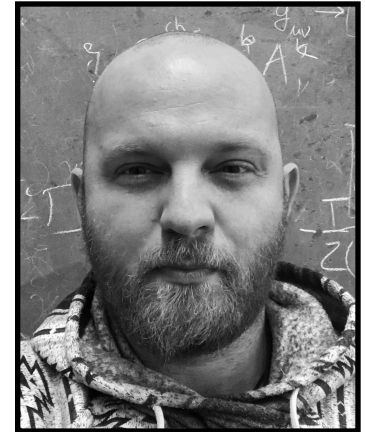
**CALEB
SCHARF**



I'll talk about how ideas on information driven states of matter play into the search for living systems in the universe. One notion is to recast this as a search for "computational zones", where computation is treated as a very general term for information driven processes and function. This has some interesting qualities, since there are energetic constraints on bitwise processes and coherence that can be mapped to environmental conditions. Another concept is the "dataome" – the externalized, seemingly symbiotic information manifested by living systems that can range from small to enormous in size and scope. The dataomes of species could have distinct observable properties that we might search for, and that might inform us about how information driven states of matter evolve.

*The Meaning of Life:
Towards an effective theory of semantic information*

**DAMIAN
SOWINSKI**



Living systems gather, process, and act on stimuli from their environments, generating novel correlations while decimating others. We, as observers of these biotic systems, can quantify the information in these observed regularities; information theory gives a plethora of tools to do so, but leaves the question of semantics unresolved: what do these observed correlations mean to the systems manipulating them? We posit an answer: *the meaning of a correlation is its use in the world*, alongside the claim that *viability* is the fundamental, intrinsic measure of utility for biotic systems. The semantic content of information to a living thing IS that information's necessity in maintaining the viability of the living thing. I go over the history of this line of thought, describe initial attempts and difficulties at formalizing a *Semantic Information Theory*, and look at how more recent attempts explore alternative operationalizations. In particular a survey of results from neural synchronization, foraging populations, and planetary biosphere models are presented.

Non-equilibrium statistical physics of computation and communication

**DAVID
WOLPERT**



In this talk I will summarize recent results on how the thermodynamics of a physical system implementing a computation depends on the computation being performed and the precise physical system being used to perform it. I will start with some new results concerning the minimal thermodynamic costs of performing a communication within a computer. Next I will summarize some results concerning the thermodynamic costs of implementing a deterministic finite automaton (DFA). After that, time allowing I will review results on how considering the minimal thermodynamic costs of computing a desired output on a Turing machine (TM), rather than the minimal size of an input string that causes the TM to produce that output (i.e., the output's Kolmogorov complexity), results in a correction term to Kolmogorov complexity. I will end by describing the vast new set of research issues at the intersection of stochastic thermodynamics and computer science theory, issues that expand both fields.

An Informational Perspective of Life

Information processing is a central feature of life and may be the key to developing biosignatures that are capable of detecting life as we do not know it. First, I will describe how concepts that we generally associate with the field of data science are strikingly descriptive of the way that life, in general, processes information about its environment. Second, I will explore how the topologies of atmospheric chemical reaction networks of Solar System bodies are distinct from one another, with potential applications to exoplanet biosignatures. Third, I will show how a robust method that combines pyrolysis–GC–MS measurements of a wide variety of terrestrial and extraterrestrial carbonaceous materials with machine-learning-based classification to achieve ~90% accuracy in the differentiation between samples of abiotic origins vs. biotic specimens. Fourth, I will propose a new time-asymmetric law that states that the functional information of a system will increase over time when subjected to selection for function(s).

**MICHAEL
WONG**





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Nicholas Marcello

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(585) 424 - 4404

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